

M2S-99: Best Practice Summary

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A definitive, actionable reference of every best practice extracted from the M2S Managed-to-SaaS Migration series (notebooks 01–08). Each practice specifies the exact setting or action, its priority, and its category. No hedging—follow these and your migration succeeds.

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1. Planning and Assessment

Practice	Setting / Action	Priority
Enforce host limit per tenant	Maximum 25,000 monitored hosts per SaaS tenant. Split into multiple tenants if consolidating Managed clusters exceeds this.	Critical
Align Managed and SaaS versions	Both environments must be on the same major version (e.g., both 1.294.x) before using SaaS Upgrade Assistant.	Critical
Require Managed cluster v1.294+	SaaS Upgrade Assistant requires Dynatrace Managed version 1.294 or later . Upgrade the Managed cluster first if needed.	Critical
Complete entity discovery before strategy	Run Entities API v2 inventory: <code>type(HOST)</code> , <code>type(SERVICE)</code> , <code>type(APPLICATION)</code> , <code>type(SYNTHETIC_TEST)</code> . Document counts in a spreadsheet.	Critical
Choose one migration tool	Pick one primary tool: SaaS Upgrade Assistant (recommended), Monaco, Terraform, or Settings API. Never mix approaches—Monaco YAML conflicts with the SaaS Upgrade Assistant.	Critical
Use SaaS Upgrade Assistant for most migrations	Install the app on the target SaaS tenant. Assign <code>upgrade-assistant:environments:write</code> IAM policy to migration users.	Recommended
Use Monaco instead for tenant consolidation	When consolidating multiple Managed tenants with identical configuration names , use Monaco—the SaaS Upgrade Assistant cannot resolve name conflicts.	Recommended
Plan for historic data loss	Historic metrics, logs, traces, problems, and session data cannot be migrated . Export critical reports before shutdown. Run dual environments to build SaaS baselines.	Critical
Coordinate licensing for dual-run	Work with Dynatrace account team to arrange temporary overlap licensing for the period both Managed and SaaS run simultaneously.	Recommended
Define measurable success criteria	Set targets: 100% host coverage, 100% service discovery, <15 min data gaps, 100% alert delivery, 100% dashboard availability, 100% integration success.	Critical
Choose migration approach by size	<500 hosts: Big Bang. 500–2,000 hosts: Phased by environment. >2,000 hosts or complex integrations: Phased by region/application.	Recommended
Engage Dynatrace Professional Services early	Involve account team and PS during the Plan phase, not during execution.	Recommended
Budget time for non-portable items	90% of configs migrate automatically; the remaining 10% (credentials, webhooks, custom scripts) takes 90% of the manual effort.	Recommended

2. Network and Architecture

Practice	Setting / Action	Priority
Open outbound 443 to SaaS endpoints	Allow HTTPS (port 443) from all monitored hosts and ActiveGates to <code>{tenant-id}.live.dynatrace.com</code> and <code>{tenant-id}.apps.dynatrace.com</code> .	Critical
Use ActiveGate routing for restricted networks	If hosts cannot reach the internet directly, deploy Environment ActiveGates. OneAgents connect to AG on port 9999 ; AG connects outbound on 443 .	Critical
Ensure DNS resolves SaaS domains	Verify DNS resolution for <code>{tenant-id}.live.dynatrace.com</code> , <code>{tenant-id}.apps.dynatrace.com</code> , and <code>*.dynatrace.com</code> .	Critical
Test connectivity before migration	Run <code>curl -v https://{tenant-id}.live.dynatrace.com/api/v1/time</code> from monitored hosts and ActiveGate servers before scheduling the migration window.	Critical
Recreate Network Zones in SaaS before migrating agents	Network Zone configuration does not transfer automatically. Create zones in SaaS via Settings > Network zones or the API, then assign AGs and OneAgents.	Critical
Match zones to physical network topology	Name zones by datacenter or region (e.g., <code>datacenter-east</code> , <code>aws-us-east-1</code>). Define alternative zones for failover.	Recommended
Deploy minimum 2 ActiveGates per network zone	Provides high availability within each zone. OneAgent auto-discovers available AGs.	Recommended
Enforce TLS 1.2+ for all traffic	Dynatrace SaaS enforces TLS 1.2 minimum. Verify no legacy TLS 1.0/1.1 configurations on proxies or load balancers.	Critical

3. ActiveGate Deployment

Practice	Setting / Action	Priority
Deploy new SaaS ActiveGates in parallel	Install new SaaS-connected AGs alongside existing Managed AGs before migrating OneAgents. Validates connectivity without impacting monitored hosts.	Critical
Size ActiveGates by host count	Up to 500 hosts: 2 CPU / 4 GB RAM / 20 GB disk . 500–1,500: 4 CPU / 8 GB / 40 GB . 1,500–5,000: 8 CPU / 16 GB / 80 GB . >5,000: scale horizontally with multiple AGs.	Recommended
Place ActiveGates close to monitored hosts	Minimize latency by deploying AGs in the same datacenter or cloud region as the hosts they serve.	Recommended
Use ActiveGate groups for role separation	Create groups: <code>production-routing</code> , <code>nonprod-routing</code> , <code>extensions</code> , <code>synthetic</code> . Assign AGs to groups based on function.	Recommended
Assign network zone during AG installation	Use <code>--set-network-zone=<zone></code> flag during installation. Post-install: add <code>networkzone=<zone></code> to <code>config.properties</code> .	Recommended
Deploy dedicated AGs for Extensions 2.0	Extensions 2.0 require host-based ActiveGate (not K8s-based). Deploy a separate AG group for extension execution.	Recommended

4. Configuration Migration

Practice	Setting / Action	Priority
Migrate configurations BEFORE redirecting OneAgents	Deploy foundational settings (management zones, tags, service detection rules) to SaaS before any agents start reporting.	Critical
Follow the 4-wave deployment order	Wave 1: Management zones, auto-tagging rules, host groups. Wave 2: Service detection, request attributes, deep monitoring. Wave 3: Alerting profiles, anomaly detection, SLOs. Wave 4: Dashboards, remaining configs.	Critical
Use SaaS Upgrade Assistant selective import	Deploy configuration types incrementally using the Smart Selective Import. Smart dependency management auto-adds required configs.	Recommended
Preview all changes before deploying	Use the SaaS Upgrade Assistant Preview Changes feature to verify edits. The app retains original values for rollback reference.	Recommended
Download deploy result CSVs for audit	After each wave, download the deployment result CSV from SaaS Upgrade Assistant. Archive these for compliance and troubleshooting.	Recommended
Manually recreate Credentials Vault entries	Cannot be exported. Re-enter all credentials (cloud platform integrations, extension credentials) directly in SaaS.	Critical
Recreate problem notification webhooks	Endpoint URLs change for SaaS. Create new Slack, Teams, PagerDuty, ServiceNow, and custom webhook integrations with SaaS URLs.	Critical
Generate new API tokens in SaaS	Managed tokens do not work in SaaS. Create new tokens with minimal required scopes for each use case.	Critical
Remove hardcoded entity IDs from dashboards	Before importing dashboards, replace hardcoded entity IDs with entity selectors. Update management zone IDs and dashboard owners.	Recommended
Recreate synthetic private locations	Deploy new Synthetic ActiveGates in SaaS. Private locations are infrastructure-specific and do not transfer.	Recommended
Update all API scripts with new base URL	Change from <code>{managed-url}/e/{env-id}</code> to <code>{tenant}.live.dynatrace.com</code> . Update tokens in all CI/CD pipelines and automation scripts.	Critical
Recreate automations in the Workflows app	Managed problem notifications become workflow triggers in SaaS. Custom webhooks become HTTP Request actions. Document existing logic first, then rebuild.	Recommended

5. OneAgent Migration

Practice	Setting / Action	Priority
Use <code>oneagentctl --set-server</code> to reconfigure (not reinstall)	Run: <code>oneagentctl --set-server="https://{tenant}.live.dynatrace.com:443/communication"</code> then <code>--set-tenant-token="{token}"</code> then <code>systemctl restart oneagent</code> . Preserves host identity and custom metadata.	Critical
Migrate non-production first, then production	Validate data flow on non-prod hosts before touching production. Apply lessons learned to each subsequent wave.	Critical
Restart application processes after OneAgent migration	Full-stack monitoring requires process restarts—Java, .NET, Node.js, PHP all inject instrumentation at process startup. Without restart: host metrics flow but no distributed tracing, no service detection, no code-level visibility .	Critical
Verify OneAgent versions are within support window	OneAgent support window is 9–12 months . Check oldest deployed versions before migration. Upgrade outdated agents first.	Recommended
Use Ansible/automation for bulk reconfiguration	For large environments, use configuration management (Ansible, Puppet, Chef) to execute <code>oneagentctl</code> commands and rolling application restarts across all hosts.	Recommended
Choose parallel install for zero-downtime requirements	Install SaaS OneAgent alongside Managed temporarily. Higher resource overhead but eliminates monitoring gap. Use only when zero-gap is mandatory.	Optional
Configure OneAgent network zone assignment	Set during reconfiguration: <code>oneagentctl --set-network-zone=<zone></code> . Must match the Network Zones created in SaaS.	Recommended
Prepare rollback procedure	Document the Managed server URL and token. Rollback: <code>oneagentctl --set-server="https://{managed-cluster}/communication"</code> + restart. Test on one host before bulk migration.	Critical

6. Security and Access Control

Practice	Setting / Action	Priority
Configure SAML SSO with full message signing	IdP must sign the entire SAML message , not just the assertion. Azure AD meets this requirement by default. Failure causes authentication errors.	Critical
Filter SAML group claims to Dynatrace groups only	Azure Entra limit: 150 groups per user in SAML claim. Filter to Dynatrace-related groups to stay under the limit.	Critical
Replace LDAP with SAML	SaaS does not support direct LDAP authentication. Migrate to SAML 2.0 SSO through your IdP.	Critical
Map Managed roles to SaaS IAM policies	Cluster admin → Account admin. Environment admin → Environment admin. Monitor user → Viewer + specific policies. Custom roles → Custom IAM policies.	Critical
Create purpose-specific API tokens with minimal scopes	Separate tokens per use case: installation (<code>InstallerDownload</code>), config migration (<code>settings.read/write</code>), queries (<code>entities.read</code> , <code>metrics.read</code> , <code>logs.read</code>), automation (<code>automation.read/write</code>).	Critical
Set API token expiration dates	Set expiration on all tokens. Store in secrets management (HashiCorp Vault, AWS Secrets Manager). Rotate regularly.	Recommended
Request SOC 2 Type II report from Dynatrace	Obtain SOC 2 Type II, ISO 27001 certificate, penetration test summaries, and Data Processing Addendum (DPA) for your security review.	Recommended
Confirm data residency region at provisioning	Select the correct region (US, EU, APAC) during tenant provisioning. Cannot be changed after provisioning.	Critical
Restrict egress to Dynatrace domains only	Firewall egress rules should allow only <code>*.dynatrace.com</code> on port 443. Block all other outbound from OneAgent and ActiveGate servers.	Recommended

7. Privacy and Data Protection

Practice	Setting / Action	Priority
Enable PII masking for request attributes	Settings → Request attributes: enable masking on all attributes capturing user input, account numbers, or personal data.	Critical
Configure session replay masking	RUM settings: mask all form inputs, text fields, and sensitive UI elements in session recordings.	Critical
Enable IP anonymization	RUM settings: enable IP anonymization to prevent storage of end-user IP addresses.	Recommended
Use OpenPipeline for log masking	Configure log processing rules in OpenPipeline to redact PII (SSNs, credit card numbers, email addresses) before storage.	Recommended
Exclude sensitive URL parameters	RUM settings: configure URL exclusion rules to strip query string parameters containing sensitive data (tokens, session IDs, PII).	Recommended
Disable session replay for internal apps	If internal apps do not require session replay, disable it to minimize data collection.	Optional
Obtain Data Processing Addendum for GDPR	Request the DPA from Dynatrace. Ensure it covers data subject access requests, right to erasure, and data portability.	Recommended

8. Validation and Post-Migration

Practice	Setting / Action	Priority
Compare entity counts: Managed inventory vs SaaS	Run <code>fetch dt.entity.host \ summarize count()</code> (and <code>service</code> , <code>application</code> , <code>process_group</code> , <code>synthetic_test</code>). Counts must match the planning-phase inventory.	Critical
Verify metrics flow with no gaps >15 minutes	Run <code>timeseries avg(dt.host.cpu.usage), from:-1h, by: {dt.entity.host}</code> and check for nulls. Any host returning null has a data gap.	Critical
Validate log ingestion is continuous	Run <code>fetch logs, from:-1h \ summarize count(), by: {bin(timestamp, 5m)}</code> and confirm no 5-minute buckets with zero count.	Critical
Confirm distributed traces are flowing	Run <code>fetch spans, from:-1h \ summarize count(), by: {bin(timestamp, 5m)}</code> . Zero spans after agent migration means application processes need restart.	Critical
Review SaaS Upgrade Assistant deploy results	Download the CSV from each deployment wave. Filter for failed configurations. Fix or exclude with documented justification.	Critical
Test alert delivery end-to-end	Trigger a test alert and confirm delivery through every notification channel (email, Slack, Teams, PagerDuty, ServiceNow).	Critical
Validate all dashboard tiles show data	Open every migrated dashboard. Tiles with "No data" indicate missing entity mappings or incorrect management zone IDs.	Recommended
Get stakeholder sign-off	Obtain written approval from: Platform Team Lead, Security Team, Application Teams, Executive Sponsor.	Critical
Keep Managed running 2–4 weeks post-migration	Maintain access for historical reference and comparison. Decommission only after full validation period.	Recommended
Archive migration artifacts	Save SaaS Upgrade Assistant deploy result CSVs, entity inventory spreadsheets, and configuration mapping documents before decommissioning Managed.	Recommended

9. Optimization and Feature Adoption

Practice	Setting / Action	Priority
Allow 7–14 days for Davis AI baselines	Response time and error rate baselines: 2–7 days . Resource usage and traffic patterns: 7–14 days . Expect higher alert volume during this period—do not disable alerting.	Critical
Tune alert thresholds after baseline period	After 2 weeks: raise thresholds on noisy alerts, add time-based conditions, lower thresholds on missing alerts.	Recommended
Optimize DQL queries on dashboards	Add explicit <code>from:</code> time filters to every query. Use <code>summarize</code> instead of fetching raw rows. Apply <code>filter</code> before aggregation. Use <code>limit</code> on all exploration queries.	Recommended
Adopt Notebooks for interactive analysis	Replace ad-hoc USQL queries with Notebook-based DQL analysis. Share notebooks across teams for collaboration. Priority: weeks 3–4 post-migration.	Recommended
Implement Workflows for automation	Replace Managed problem notifications with SaaS Workflow triggers + HTTP Request actions. Priority: weeks 5–6 post-migration.	Recommended
Explore Davis Copilot for AI-assisted analysis	Enable Davis Copilot for natural language querying and root cause analysis. Priority: weeks 7–8 post-migration.	Optional
Right-size data retention for cost optimization	Configure Grail bucket retention by data type. Logs: set retention based on compliance needs. Metrics: leverage the 5-year aggregated retention. Use <code>bucket:</code> targeting in queries to limit scan scope.	Recommended
Update all runbooks and documentation	Replace Managed URLs, procedures, and screenshots with SaaS equivalents. Update architecture diagrams, onboarding guides, and disaster recovery plans.	Recommended

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